

Question Number	Answer	Additional guidance	Mark
8(a)(i)	• Z		(1)

Question Number	Answer	Additional guidance	Mark
8(a)(ii)	• cortical granule(s)	Mark 1 st answer accept cortical vesicle accept phonetic spellings ignore contractile granules ignore cortical enzymes	(1)

Question Number	Answer	Mark
8(b)(i)	The only correct answer is B one <i>A is not correct because the acrosome is a membrane bound sac containing digestive enzymes</i> <i>C is not correct because mitochondria perform aerobic respiration and the nucleus contains linear DNA</i> <i>D is not correct because mitochondria perform aerobic respiration and the nucleus contains linear DNA</i>	(1)

Question Number	Answer	Additional guidance	Mark
8(b)(ii)	4.48 : 1 / 4.5:1	accept answer in range 4.3 to 4.7 ignore 4:1	(1)

Question Number	Answer	Additional guidance	Mark
8(b)(iii)	An answer that makes reference to the following points: - the (mean) swimming speed (of sperm cells) increases as width of helical membrane increases (1) Reed bunting males (need sperm cells with faster swimming speeds as they) compete with other males / the female Reed bunting mates with more than one male (1) {faster swimming sperm cells / larger helical membranes} are more likely to {fertilise egg cells / form a zygote / have their alleles inherited} (before other males' sperm) (1) - males with larger acrosome (to nucleus ratio are more likely to) {have a quicker acrosome reaction / digest the zona pellucida (more) quickly} (1)	max 3 without reference to reed bunting or nuthatch accept swimming speed increases as membrane increases / converse accept wide helical membrane increases swimming speed / converse accept Reed bunting sperm cells will have a fast (mean) swimming speed / converse for Nuthatch accept converse for Nuthatch accept converse accept selection pressure linked to either graph /table info ignore reaching egg cell first accept larger acrosome (to nucleus ratio) have more digestive enzymes	(4)

Question Number	Answer	Additional guidance	Mark
8(c)	An answer that makes reference to the following points: • can reproduce with any blue dragon they encounter / in any one encounter both can have fertilised eggs (1) • increased number of {offspring produced / population} (in a certain time period) (1) IMP • (sexual reproduction) increases genetic {variation / diversity} (1)	ignore can mate with both males and females accept both can be fertilised / higher chance of {finding a mate / reproduction} ignore self-fertilisation accept increased {chance of fertilisation / reproductive success / fertilised gametes} accept low risk of extinction / increase chance of survival (of species) do not award mp2 in context of self-fertilisation accept (sexual reproduction) allows genetic {diversity / variation} accept genetic variation means less susceptible to environmental changes ignore increased biodiversity unqualified ignore inbreeding	(3)

Question Number	Answer	Additional guidance	Mark
8(d)	• calculation of q (1) • calculation of number of cats with genotype BB (1) • calculation of number of cats with genotype Bb (1)	$\sqrt{0.16} = 0.4$ (or if you see 0.6 in working) $(0.6^2) \times 100 = 36$ $(2 \times 0.4 \times 0.6) \times 100 = 48$ both correct cat numbers scores full marks no ecf	(3)